

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DEEKSHITHA U
Register Number	192424055
Date	29 September 2026
Day	Tuesday

SET 2: CUSTOMER FEEDBACK ANALYSIS

Aim

To analyze customer satisfaction data using a histogram, pie chart, stacked bar chart, and word cloud to understand customer age distribution, satisfaction scores, age-group-wise satisfaction, and common feedback terms.

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1: Histogram of Customer Ages

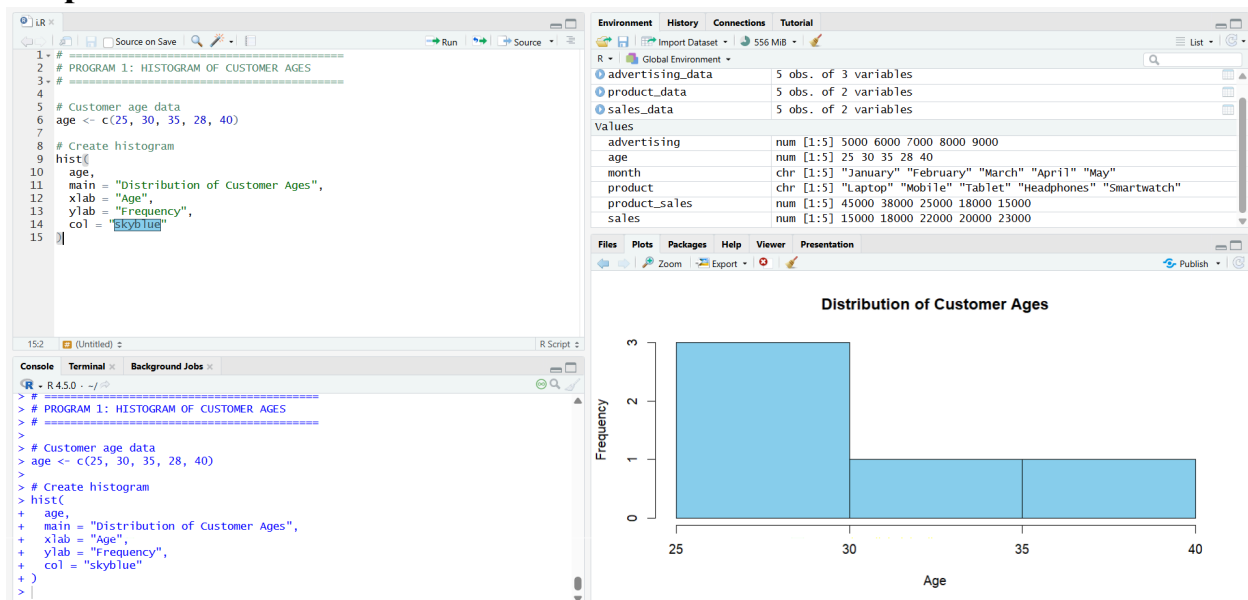
Question

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

R Program

```
# =====  
# PROGRAM 1: HISTOGRAM OF CUSTOMER AGES  
# =====  
# Customer age data  
age <- c(25, 30, 35, 28, 40)  
# Create histogram  
hist(  
  age,  
  main = "Distribution of Customer Ages",  
  xlab = "Age",  
  ylab = "Frequency",  
  col = "skyblue"  
)
```

Output



Observation

The histogram represents the distribution of customer ages in the given dataset.

PROGRAM 2: Pie Chart of Satisfaction Scores

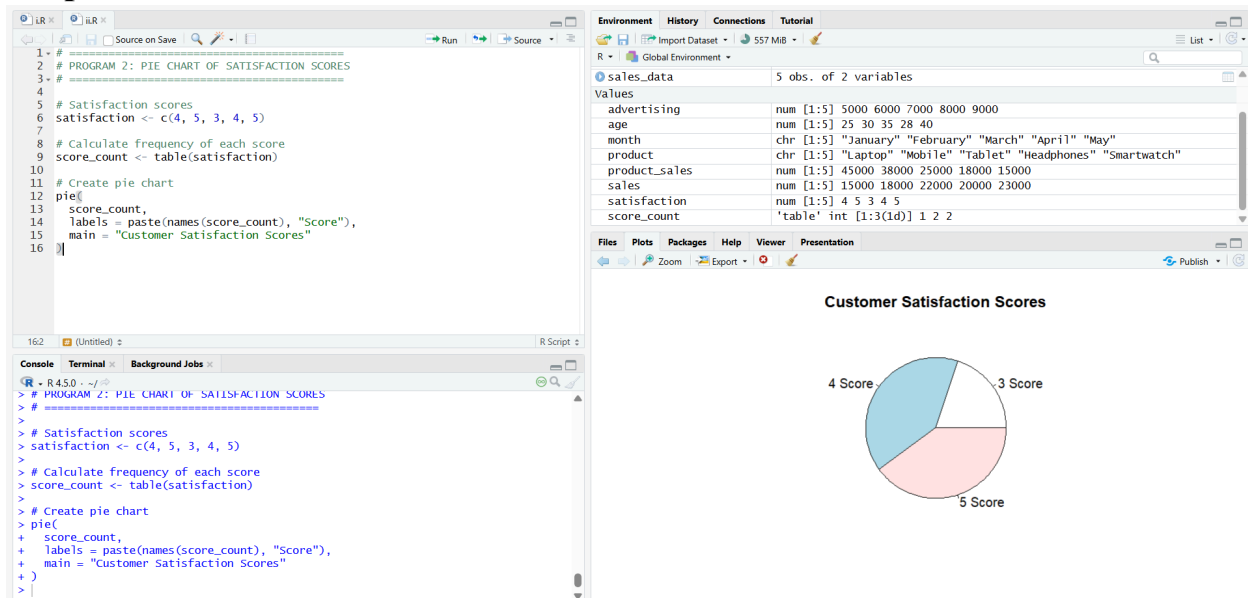
Question

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

R Program

```
# =====  
# PROGRAM 2: PIE CHART OF SATISFACTION SCORES  
# =====  
# Satisfaction scores  
satisfaction <- c(4, 5, 3, 4, 5)  
# Calculate frequency of each score  
score_count <- table(satisfaction)  
# Create pie chart  
pie(  
  score_count,  
  labels = paste(names(score_count), "Score"),  
  main = "Customer Satisfaction Scores"  
)
```

Output



Observation

Satisfaction scores 4 and 5 each occur twice, while satisfaction score 3 occurs once.

PROGRAM 3: Stacked Bar Chart by Age Group

Question

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

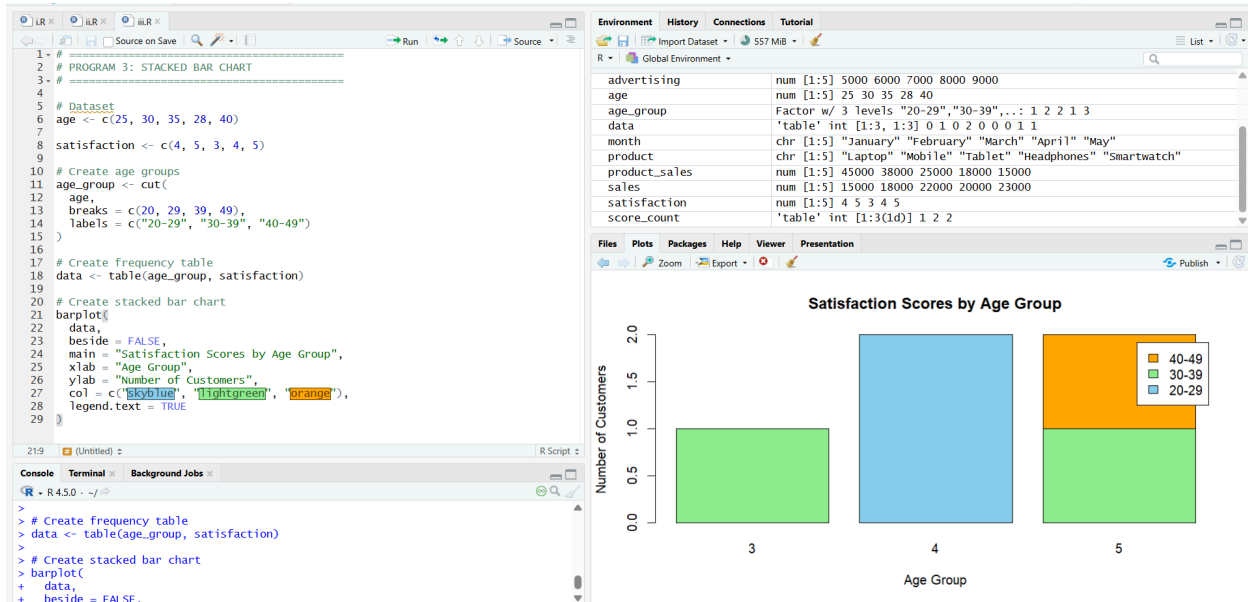
We first divide the customers into three age groups:

- 20–29
- 30–39
- 40–49

R Program

```
# =====  
# PROGRAM 3: STACKED BAR CHART  
# =====  
  
# Dataset  
age <- c(25, 30, 35, 28, 40)  
  
satisfaction <- c(4, 5, 3, 4, 5)  
  
# Create age groups  
age_group <- cut(  
  age,  
  breaks = c(20, 29, 39, 49),  
  labels = c("20-29", "30-39", "40-49")  
)  
  
# Create frequency table  
data <- table(age_group, satisfaction)  
  
# Create stacked bar chart  
barplot(  
  data,  
  beside = FALSE,  
  main = "Satisfaction Scores by Age Group",  
  xlab = "Age Group",  
  ylab = "Number of Customers",  
  col = c("skyblue", "lightgreen", "orange"),  
  legend.text = TRUE  
)
```

Output



Observation

The stacked bar chart represents the distribution of satisfaction scores across the three age groups.

PROGRAM 4: Word Cloud of Customer Feedback

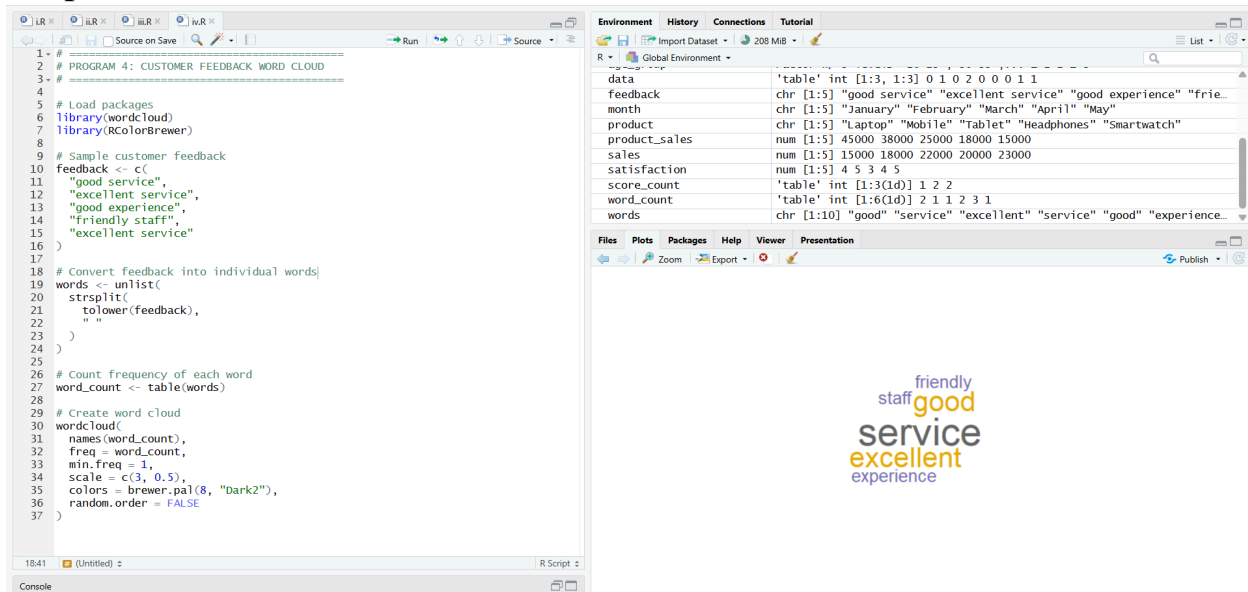
Question

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

R Program

```
# =====  
# PROGRAM 4: CUSTOMER FEEDBACK WORD CLOUD  
# =====  
  
# Load packages  
library(wordcloud)  
library(RColorBrewer)  
# Sample customer feedback  
feedback <- c(  
  "good service",  
  "excellent service",  
  "good experience",  
  "friendly staff",  
  "excellent service"  
)  
# Convert feedback into individual words  
words <- unlist(  
  strsplit(  
    tolower(feedback),  
    " "  
  )  
)  
# Count frequency of each word  
word_count <- table(words)  
  
# Create word cloud  
wordcloud(  
  names(word_count),  
  freq = word_count,  
  min.freq = 1,  
  scale = c(3, 0.5),  
  colors = brewer.pal(8, "Dark2"),  
  random.order = FALSE  
)
```

Output



Observation

The word cloud highlights frequently occurring terms in the supplied sample feedback, including service, excellent, good, friendly, and experience.

Procedure

1. Create the customer age and satisfaction score vectors in R.
2. Use `hist()` to represent the distribution of customer ages.
3. Use `table()` and `pie()` to display the satisfaction score distribution.
4. Create age groups using `cut()`.
5. Create a frequency table using `table()`.
6. Use `barplot()` to create the stacked bar chart.
7. Install and load the `wordcloud` and `RColorBrewer` packages.
8. Prepare sample open-ended customer feedback.
9. Split the feedback into individual words.
10. Count the frequency of each word.
11. Use `wordcloud()` to visualize the frequently occurring words.

Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, and stacked bar chart. A word cloud was also created from sample customer feedback to identify frequently occurring feedback terms.